Signals and Systems time domain representation and Signal Operations

Participated in & did the coursework [Y/N]?

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1 Introduction

Representing signals in the time domain is essential for analyzing how systems respond to external inputs in continuous-time signal processing [1]. Aligned with that, the primary objective of this laboratory exercise is to mathematically define, manipulate, and visualize continuous-time signals and systems. Where the experiment aims to apply fundamental time-domain operations including time shifting, time flipping, and time scaling to given piecewise sequences. The exercise also aims to determine and plot the impulse responses of ideal frequency-selective filters (lowpass, highpass, bandpass, and bandstop) and evaluate the auto-correlation of a given signal to identify its temporal characteristics.

These objectives are grounded in the fundamental theory of linear time-invariant (LTI) systems, which can be analyzed through three interrelated concepts: temporal manipulation, system characterization, and internal signal analysis. First, regarding temporal manipulation, time-domain operations fundamentally alter a signal's independent variable [1]. Specifically, time shifting advances or delays the waveform along the time axis, time flipping reflects it across the origin, and time scaling compresses or expands its duration [2]. Second, regarding system characterization, an LTI system's behavior is entirely defined by its impulse response, which dictates the system's reaction to an instantaneous unit impulse [3]. Evaluating these responses for specific filters yields critical insight into their transient behavior and frequency selectivity [3]. Third, regarding internal signal analysis, comprehensive evaluation necessitates examining a signal's inherent properties independent of external filters [4]. Rather than observing how a signal interacts with a system, analysts must observe how a signal interacts with itself over time. Auto-correlation provides the exact mathematical mechanism for this internal comparison, measuring the similarity between a waveform and its own time-delayed counterpart to detect hidden periodicities or structural lags [4].

To computationally achieve these theoretical objectives, a specific, organized suite of MATLAB functions is required. For precise mathematical modeling, the syms function is utilized to declare symbolic variables. For signal construction, piecewise continuous sequences and system excitations are generated using the heaviside (unit step) and dirac (unit impulse) commands, which are subsequently augmented by exp and cos to model exponential decay and sinusoidal oscillations. For statistical computation, the xcorr function is used to calculate the discrete auto-correlation sequence. Finally, for data visualization, figure, subplot, and plot are employed to generate multi-axis graphs that seamlessly compare the manipulated signal sequences and filter responses.

2 Experimental Design

The experimental design is structured to systematically investigate the properties of continuous-time signals through mathematical modeling and computational simulation. By leveraging MATLAB's Symbolic Math Toolbox, this methodology demonstrates how theoretical signal operations can be implemented and validated within a software environment like MATLAB.

2.1 Signal Construction and Transformation Methodology

The core of this experiment involves the symbolic definition of a baseline signal, $x(t)$, which serves as the primary subject for various time-domain operations. The signal is constructed using Heaviside step functions to isolate its behavior within the specific interval of $t = 0$ to $t = 8$ seconds. The exact symbolic math expression designed for this experiment is:

$$x(t) = (1 - 0.25t) \cdot (\text{heaviside}(t) - \text{heaviside}(t - 8)) \quad (1)$$

The experimental procedure for signal transformation required the modification of the independent variable t to achieve specific outcomes:

- **Time Shifting**: Replacing t with $(t-2)$ for a delay and $(t+4)$ for an advance.
- **Time Reversal**: Substituting t with $(-t)$.

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- **Combined Operations**: Implementing $x(3-t)$ to achieve both flipping and shifting.
- **Time Scaling**: Multiplying t by a constant factor of 2 ($x(2t)$) to achieve compression.

These transformations were consolidated into a single figure using a 3×2 subplot configuration to allow for simultaneous comparative analysis of the six resulting waveforms.

2.2 Impulse Response Modeling for Filter Systems

The experiment also addressed the representation of system characteristics through the definition of impulse response functions ($h(t)$). Four distinct filter types were modeled using combinations of the Dirac delta function ($\delta(t)$) and the Heaviside step function ($u(t)$) to simulate ideal signal processing behavior. The specific equations implemented for these models are summarized in Table 1.

Table 1
Mathematical Expressions for Filter Impulse Responses

Filter Type	Symbolic Equation	MATLAB Implementation
Lowpass	$h_1(t) = e^{-10t}u(t)$	`exp(-10*t) * heaviside(t)`
Highpass	$h_2(t) = \delta(t) - e^{-10t}u(t)$	`dirac(t) - exp(-10*t) * heaviside(t)`
Bandpass	$h_3(t) = e^{-10t}\cos(500t)u(t)$	`exp(-10*t)*cos(500*t)*heaviside(t)`
Bandstop	$h_4(t) = \delta(t) - e^{-10t}\cos(500t)u(t)$	`dirac(t) - exp(-10*t)*cos(500*t)*heaviside(t)`

2.3 Numerical Computation and Statistical Correlation

To determine the auto-correlation of the signal $x(t)$, the experiment required a transition from symbolic expressions to numerical data arrays. This was achieved by defining a discrete time vector ranging from -10 to 10 seconds with a high-resolution step size of $dt = 0.01$. The symbolic expression $x(t)$ was evaluated over this vector to create the numerical array `x_num`. The auto-correlation was computed using the `xcorr` function, and the resulting sequence was normalized by multiplying by the time step dt.

2.4 Experimental Pseudocode

The following logic describes the programmatic implementation of the experiment:

1) START
2) DEFINE symbolic variable 't'
3) CONSTRUCT signal x(t) using Heaviside functions for interval [0, 8]
4) FOR EACH transformation (shift, flip, scale):
   - APPLY transformation to t
   - PLOT result in 3x2 subplot grid (Figure 1)
5) CONSTRUCT signal y(t) = exp(-t)u(t) + u(-1-t)
6) PLOT y(t) for t in range [-6, 6] (Figure 2)
7) DEFINE h1(t) through h4(t) for LPF, HPF, BPF, and BSF
8) OVERLAY h1(t) and h2(t) (excluding delta) on same axis (Figure 3)
9) CONVERT symbolic x(t) to numeric array x_num with dt = 0.01
10) COMPUTE Rxx = xcorr(x_num, x_num) * dt
11) PLOT Rxx vs tau (Figure 4)
12) END

3 Analysis of Results

The results obtained from the lab activity provide a comprehensive look at the mathematical manipulation of signals and the resulting waveforms. This section consolidates the findings to demonstrate the practical application of signal operations and system modeling in a computational environment.

3.1 Time-Domain Transformations of the Baseline Signal

The first phase of the analysis focused on the manipulation of the baseline signal $x(t)$, which is a linear ramp function defined within the interval $t \in [0, 8]$. As observed in the consolidated results, the original signal transitions from an amplitude of 1 at $t = 0$ to an amplitude of -1 at $t = 8$. The transformation $x(t-2)$ demonstrated a successful time delay, where the entire waveform shifted two units to the right without altering its shape or amplitude. Conversely, $x(t+4)$ showed a time advance, moving the signal four units to the left into the negative time domain. The time-reversal operation $x(-t)$ resulted in a vertical reflection across the y-axis, effectively flipping the ramp's orientation. The combined operation $x(3-t)$ proved to be the most complex, as it required both a reversal and a 3-second shift, resulting in a signal that peaks at $t = 3$. Finally, the time-scaling operation $x(2t)$ compressed the original 8-second duration into a 4-second window, confirming that a scaling factor greater than unity increases the "speed" of the signal relative to the time axis.

Placeholder: [Figure 1. Signal Transformations: x(t), x(t-2), x(t+4), x(-t), x(3-

Figure 1. Subplots of Signal Transformations illustrating shifting, reversal, and scaling.

3.2 Piecewise Evaluation of Composite Signal $y(t)$

The composite signal $y(t) = e^{-t}u(t) + u(-1-t)$ was analyzed to observe the interaction between decaying and constant signal components. The resulting plot reveals three distinct operational regions that align with theoretical expectations. For $t < -1$, the signal maintains a constant unit amplitude of 1, governed by the time-reversed and shifted unit step function $u(-1-t)$. A critical observation is the "dead zone" or gap between $t = -1$ and $t = 0$, where neither part of the piecewise function is active, resulting in an amplitude of exactly zero. At $t = 0$, the signal abruptly returns to an amplitude of 1 and immediately begins a smooth exponential decay toward zero, as dictated by the $e^{-t}u(t)$ term. This simulation confirms the system's ability to model signals with high-frequency discontinuities and differing temporal characteristics in a single expression.

Placeholder: [Figure 2. Signal y(t) showing constant segments, a zero-amp

Figure 2. Visualization of composite signal y(t) highlighting piecewise discontinuities.

3.3 Complementary Impulse Responses of Filter Systems

The comparative analysis of the impulse responses for the lowpass and highpass filters highlights the mathematical relationship between different system types. The lowpass response $h_1(t)$ is characterized by a positive exponential decay that starts at 1 and approaches zero as time increases, which effectively models the attenuation of high-frequency components. In contrast, the highpass response $h_2(t)$—when plotted excluding the initial Dirac delta impulse—starts at a negative amplitude of -1 and decays upward toward the zero-axis. This inverse relationship is expected, as the highpass filter in this experiment is mathematically derived by subtracting the lowpass response from the input impulse. The visualization confirms that both filters reach a steady state of zero within approximately 0.6 seconds, indicating a stable system response.

Placeholder: [Figure 3. Comparison of Lowpass and Highpass Impulse Responses over 1 second]

Figure 3. Overlay of filter impulse responses demonstrating complementary decay behaviors.

3.4 Statistical Evaluation of Auto-correlation $R_{xx}(\tau)$

The auto-correlation plot provides a statistical measure of how much the signal $x(t)$ overlaps with itself at various time lags. The resulting waveform is perfectly symmetric around the vertical axis $\tau = 0$, where the peak amplitude of approximately 2.7 occurs. This peak represents the total energy of the signal. As the lag τ increases or decreases, the correlation values transition from positive to negative, specifically forming two distinct "lobes" that reach a minimum of approximately -1.1. These negative values occur because the ramp signal $x(t)$ contains both positive and negative amplitudes; when shifted by certain lags, the positive portions of the original signal align with the negative portions of the shifted version, resulting in a negative correlation product. The correlation eventually returns to zero when the lag exceeds ± 8 seconds, as there is no longer any overlap between the two versions of the signal.

Placeholder: [Figure 4. Auto-correlation Rxx(tau) showing symmetry and energy distribution]

Figure 4. Auto-correlation function highlighting temporal symmetry and peak energy at zero lag.

3.5 Numerical Results and Dataset Characteristics

To ensure the accuracy of the simulations, specific numerical parameters were maintained across all group members' experiments. The transition from symbolic expressions to numerical data utilized a sampling interval of $dt = 0.01$ seconds, which equates to a sampling frequency of 100 Hz. This resolution was sufficient to capture the rapid transitions in the filter impulse responses and the slope of the ramp function without significant aliasing. The numerical array `x_num` was generated over a 20-second window, resulting in a vector of 2001 data points. Consequently, the auto-correlation result `Rxx` produced a vector of 4001 points, adhering to the standard signal processing rule where the length of a correlation or convolution is $2N - 1$. The final amplitude of the auto-correlation peak (2.7) was achieved by multiplying the raw `xcorr` output by the time step dt, which correctly approximates the continuous-time integral of the signal's square.

Table 2
Consolidated Numerical Parameters and Simulation Results

Parameter	**Group Consensus Value**	**Description**
Sampling Interval (dt)	0.01	Time resolution for numerical arrays
Time Range	-10 to 10 seconds	Total window for signal observation
Input Vector Size	2001 samples	Dimensions of `x_num`
Correlation Vector Size	4001 samples	Dimensions of `Rxx`
Max Correlation ($R_{xx}(0)$)	2.7	Peak energy value at zero lag

4 Discussion and Conclusion

The laboratory exercise successfully demonstrated the mathematical modeling, manipulation, and characterization of continuous-time signals and systems using MATLAB, fulfilling the primary objective to define and visualize these sequences. The first phase of the experiment confirmed that temporal manipulations, specifically time shifting, flipping, and scaling, directly alter a signal's independent variable to achieve predictable geometric changes. As evidenced in the results, the baseline signal $x(t)$, which originally spanned from $t = 0$ to $t = 8$ seconds, was successfully shifted, reflected, and compressed; for instance, the operation $x(2t)$ halved the signal duration to 4 seconds. These transformations validated the foundational theory of Linear Time-Invariant (LTI) systems by showing how signal geometry is modified through precise mathematical operations.

Furthermore, the characterization of systems through impulse responses $h(t)$ provided critical insight into filter behavior and frequency selectivity. By implementing symbolic equations for lowpass, highpass, bandpass, and bandstop filters, the experiment showcased how different systems react to an instantaneous unit impulse. A direct comparison between the lowpass and highpass responses revealed a notable symmetry: the lowpass response $h_1(t)$ exhibited a positive exponential decay starting at an amplitude of 1, while the highpass response $h_2(t)$, excluding the initial impulse, mirrored this with a negative decay starting at -1. This practical application of the Dirac delta $\delta(t)$ and Heaviside step $u(t)$ functions in MATLAB reinforced the theoretical understanding of how LTI systems are entirely defined by their impulse responses.

Finally, the experiment explored internal signal analysis through the computation of auto-correlation $R_{xx}(\tau)$, which measured the signal's temporal similarity to itself. By transitioning from symbolic expressions to numerical data arrays with a high-resolution time step of $\Delta t = 0.01$, the total energy of the signal was identified at the zero-lag peak, reaching an amplitude of approximately 2.7. The resulting symmetric plot confirmed that as the time lag τ increases, the correlation drops, eventually returning to zero when the waveforms no longer overlap. In conclusion, the integration of MATLAB's computational tools effectively bridged theoretical concepts with practical simulation, providing a robust framework for analyzing the interaction between signals and systems.

5 Authors Contribution

On what contributions to specify, see the terms at www.elsevier.com.

1) BAUTISTA, Alyssa Nicole : Methodology, Software, Validation, Formal Analysis, Writing – Original Draft, and Review and Editing, Visualization, Project Administration
2) GONZALES, Jeremia E.: Conceptualization, Software, Investigation, Resources, Data Curation, Writing – Original Draft, and Review and Editing, Supervision
3) MENDIOLA, Keifer Judd: Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, and Review and Editing

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